MISSINGNESS-RELIABLE REGIONAL GRAPH FORECASTING FOR SPATIAL TIME SERIES

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ABSTRACT

Missing observations in spatial time series not only shorten temporal context but also make learned sensor relations unreliable. We propose Missingness-Reliable Multi-Scale Graph Forecasting (MR-MSGF), a direct forecasting model that predicts from incomplete histories without a separately supervised imputation stage. MR-MSGF builds a window-specific relation graph whose learned similarities are regularized by pairwise co-observation, fuses it with physical topology, and coarsens the fused relation into physically connected regions. A lightweight reliability gate then controls when regional context should influence multi-scale prediction. Experiments on METR-LA show that MR-MSGF improves forecasting under medium-to-high contiguous and point-wise missing rates, demonstrating the value of co-observation-aware relation learning for incomplete spatial histories.

Index Terms— Multivariate time series forecasting, missing observations, spatiotemporal graph learning, graph coarsening

1. INTRODUCTION

Traffic, environmental, and industrial forecasting systems depend on both temporal context and inter-sensor relations. Spatiotemporal graph models commonly use physical topology or learned adjacency to represent such relations [1, 2], and hybrid designs show that physical and data-driven dependencies can be complementary [3]. In real sensor networks, however, missing observations caused by faults or communication interruptions weaken two information sources at once: the observed temporal history of each sensor and the evidence used to infer pairwise relations.

A common solution is to impute missing history before forecasting, but unsupported reconstructions may introduce errors that are then propagated to prediction. Direct forecasting methods instead encode values and masks jointly [4, 5]. For graph forecasting under missingness, BiTGraph injects missing-pattern biases into temporal and graph propagation [6], while GinAR combines predefined and adaptive relations when variable missingness damages dependency estimation [7]. Hierarchical models such as HD-TTS and P-GUTS further aggregate information across spatial and temporal resolutions [8, 9]. These advances improve missing-aware representations, yet they rarely ask whether a learned edge in the current window is supported by enough joint observations, or whether regional context should be trusted for a particular node.

This paper addresses the reliability of relations rather than historical reconstruction. The key idea is simple: an edge inferred from two nodes that were rarely observed together should not be treated the same as an edge supported by dense co-observation. We therefore propose Missingness-Reliable Multi-Scale Graph Forecasting (MR-MSGF), which directly forecasts from incomplete spatial history and regulates fine-to-coarse message passing by observation evidence. The contributions are:

- We construct a window-specific observation-supported graph that combines learned similarity with pairwise co-observation evidence.
- We fuse the observation-supported graph with a physical graph before projecting the relation to physically connected regions.
- We use node-, edge-, and region-level reliability statistics to gate regional residual messages and bias multi-scale readout.

Unlike imputation-first pipelines, MR-MSGF does not ask the model to reconstruct missing history before prediction. Unlike adaptive graph methods that learn a relation from incomplete values alone, it exposes the amount of joint observation that supports each candidate edge. This distinction is important in missing-history forecasting because the same learned similarity can have different reliability depending on how much of the two histories was actually observed.

2. METHOD

2.1. Problem Setup and Backbone

Let $X \in \mathbb{R}^{T\times N\times C}$ be a single history window from N sensors and let $M \in \{0,1\}^{T\times N\times C}$ indicate observed entries. The model receives $X^{\text{obs}} = M \odot X$, M, and a physical adjacency $A^{\text{phy}} \in \mathbb{R}^{N\times N}$, and predicts $Y \in \mathbb{R}^{H\times N\times C_y}$. In a mini-batch, relation graphs and reliability statistics are computed independently for each history window; we omit the batch index below. For graph construction, we use the channel-averaged mask $\bar{M}_{t,i} = C^{-1}\sum_c M_{t,i,c}$ and define

$$r_i = \frac{1}{T}\sum_t \bar{M}_{t,i}, \qquad q_{ij} = \frac{1}{T}\sum_t \bar{M}_{t,i}\bar{M}_{t,j}. \tag{1}$$

Here r_i is node availability and q_{ij} is pairwise co-observation, and $\text{RowNorm}(A)_{ij} = A_{ij}/(\sum_j A_{ij} + \epsilon)$. To distinguish the proposed relation modeling from the existing temporal representation, a mask-aware encoder and a dilated recurrent hierarchy first produce $H_\ell \in \mathbb{R}^{N\times d}$, which is passed to the original spatial hierarchy to obtain multi-resolution states $B_{s,\ell}$. Here, ℓ indexes temporal levels, $s = 0$ denotes the finest spatial level, and $s > 0$ denotes the original coarser spatial levels. This hierarchy adopts established missingness-aware temporal modeling and multi-scale readout [8]; our relation branch replaces only $B_{0,\ell}$, while all $B_{s,\ell}$ for $s > 0$ remain available to the readout.

2.2. Observation-Supported Relation Fusion

The encoder produces history features $E \in \mathbb{R}^{T\times N\times d}$. We obtain a window-specific node representation by masked pooling:

$$u_i = \frac{\sum_t \bar{M}_{t,i}E_{t,i}}{\sum_t \bar{M}_{t,i} + \epsilon}. \tag{2}$$

The observation-supported score combines learned similarity and co-observation:

$$s_{ij} = \frac{(W_q u_i)^\top (W_k u_j)}{\sqrt{d}}, \tag{3}$$

$$\tilde{s}_{ij} = s_{ij} + \text{softplus}(\beta)\log(q_{ij} + \epsilon). \tag{4}$$

Here β is a learnable scalar. For each node, we retain the top-k entries of $\tilde{s}_{ij}$, apply a row softmax, add self-loops, and row-normalize to obtain a window-specific directed sparse graph A^{obs}; we set $k = 8$ in all experiments. The logarithmic term is a soft evidence penalty: an edge with little joint observation is discouraged before sparsification, without imposing a hard correlation rule.

Let $\bar{A}^{\text{phy}}$ denote the physical graph after adding self-loops and applying row normalization. A learned edge gate then forms the fine relation:

$$g_{ij} = \text{sigmoid}\Big(\psi_g([r_i, r_j, q_{ij}, \bar{A}^{\text{phy}}_{ij}, A^{\text{obs}}_{ij}])\Big), \tag{5}$$

$$A^{\text{fine}} = \text{RowNorm}\big(g \odot \bar{A}^{\text{phy}} + (1-g)\odot A^{\text{obs}}\big). \tag{6}$$

Thus, both physical structure and observation-supported relations participate in end-to-end learned relation fusion. The gate is intentionally edge-specific rather than a global mixing coefficient. When observation support is scarce, physical adjacency supplies stable structural information; with sufficient joint observations, an observation-supported edge can express dynamic dependence beyond physical proximity. In implementation, ψ_g is a two-layer $5\to16\to1$ MLP with GELU activation.

2.3. Reliability-Controlled Regional Context

We construct a fixed hard assignment $S \in \{0,1\}^{N\times K}$ only from the physical graph, with each node assigned to exactly one region. We symmetrize the physical adjacency and remove self-loops, seed each region with the highest remaining physical-degree node, and greedily add the unassigned node with the largest positive affinity to any region member. A region stops growing if no such node exists or it reaches its maximum size m; we use $m = 8$. This construction guarantees physical connectivity and does not group zero-affinity nodes merely to fill a region. With $n_k = \sum_i S_{ik}$, the fine state, regional state, and coarse relation are

$$Z^{\text{fine}}_\ell = \text{ELU}(A^{\text{fine}}H_\ell W_f + H_\ell W_s), \tag{7}$$

$$H^{\text{reg}}_{\ell,k} = \frac{1}{n_k}\sum_i S_{ik}Z^{\text{fine}}_{\ell,i}, \tag{8}$$

$$A^{\text{coarse}} = \text{RowNorm}(S^\top A^{\text{fine}} S). \tag{9}$$

Regional propagation and lifting are

$$Z^{\text{reg}}_\ell = \text{ELU}(A^{\text{coarse}}H^{\text{reg}}_\ell W_c + H^{\text{reg}}_\ell W_r), \tag{10}$$

$$Z^{\text{coarse}}_\ell = SZ^{\text{reg}}_\ell. \tag{11}$$

Coarsening after fine-scale fusion lets regional aggregation operate on a graph containing both physical structure and observation-supported relations; coarsening only the physical or observation graph cannot retain both sources of evidence.

For node i, let $\bar{q}_i = \sum_j \bar{A}^{\text{phy}}_{ij}q_{ij}$ be physical-support-weighted co-observation, $\rho_i = \sum_k S_{ik}n_k^{-1}\sum_j S_{jk}r_j$ be regional coverage, and b_i be the longest missing block in the current history window divided by T. Let $k(i)$ denote the region of node i, $c_{i,\ell} = H^{\text{reg}}_{\ell,k(i)}$, and

$$\zeta_{i,\ell} = \sqrt{\max\Big(d^{-1}\|Z^{\text{fine}}_{\ell,i} - c_{i,\ell}\|_2^2, \epsilon\Big)}, \tag{12}$$

$$\delta_{i,\ell} = \frac{\zeta_{i,\ell}}{N^{-1}\sum_j \zeta_{j,\ell} + \epsilon}. \tag{13}$$

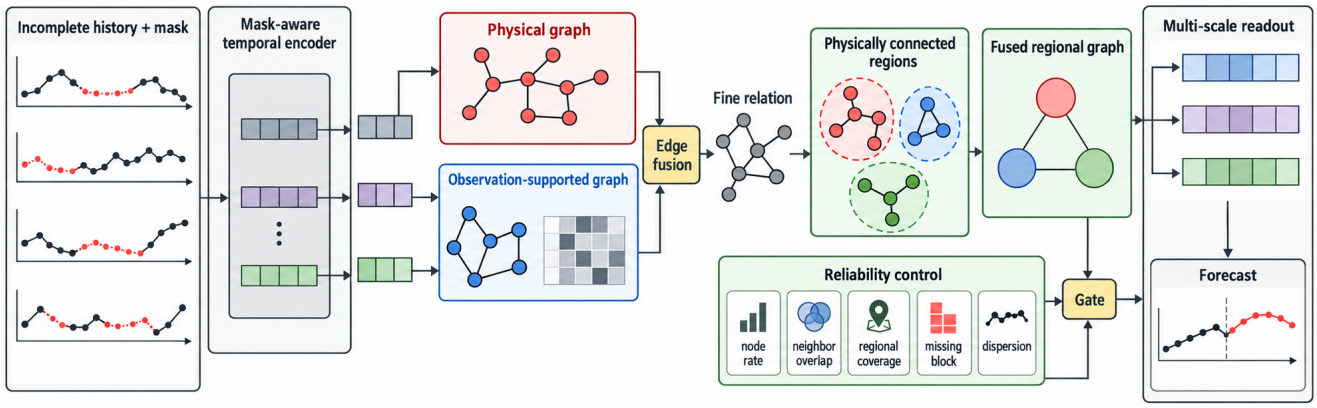


Fig. 1. Overview of MR-MSGF. A mask-aware encoder forms temporal states from incomplete history. A co-observation-supported graph is fused with the physical graph before physically connected regional coarsening. Reliability evidence regulates regional residual injection and multi-scale readout.

Here $\delta_{i,\ell}$ is normalized representation dispersion from the regional center. The regional gate is

$$\alpha_{i,\ell} = \text{sigmoid}\left(\psi_\alpha([r_i, \bar{q}_i, \rho_i, b_i, \delta_{i,\ell}]) \right. \tag{14}$$

$$\left. + \gamma^\top [1 - r_i, 1 - \bar{q}_i, \rho_i - 1, b_i, -\delta_{i,\ell}]\right). \tag{15}$$

Here ψ_α uses the same two-layer MLP structure as the edge gate, and γ is a learnable scaling vector. The second term supplies missingness patterns and regional consistency as an additional residual signal. Since γ is not sign-constrained, we do not interpret this term as a monotonic prior. Let $\boldsymbol{\alpha}_\ell = [\alpha_{1,\ell}, \ldots, \alpha_{N,\ell}]^\top \in \mathbb{R}^N$, broadcast along the feature dimension. The finest representation becomes

$$\widetilde{B}_{0,\ell} = Z_\ell^{\text{fine}} + \boldsymbol{\alpha}_\ell \odot \left(Z_\ell^{\text{coarse}} W_p\right), \tag{16}$$

while $\widetilde{B}_{s,\ell} = B_{s,\ell}$ for $s > 0$. We further bias multi-scale readout by reliability evidence. Let $R_i = [r_i, \bar{q}_i, \rho_i, b_i]^\top$. The score for prediction step h is

$$e_{h,s,\ell,i} = e^{\text{repr}}_{h,s,\ell,i} + w_{s,\ell}^\top R_i. \tag{17}$$

For each node and prediction step, a softmax over (s, ℓ) yields $\pi_{h,s,\ell,i}$, and $F_{h,i} = \sum_{s,\ell} \pi_{h,s,\ell,i} \widetilde{B}_{s,\ell,i}$ is mapped to $\hat{Y}_{h,i}$ by the prediction head. Thus, reliability statistics bias scale selection rather than replacing representation content. Training minimizes masked forecasting MAE,

$$\mathcal{L}_{\text{pred}} = \frac{\sum M^Y \odot |\hat{Y} - Y|}{\sum M^Y}, \tag{18}$$

where M^Y excludes unavailable future targets. We do not use reconstruction, graph, or clustering auxiliary losses. The gate is not interpreted as calibrated uncertainty. It is a forecasting-oriented control variable that decides whether regional context should be added for a node at a temporal level. This design keeps the added branch lightweight while making regional messages conditional on missingness evidence rather than unconditional smoothing.

3. EXPERIMENTS

3.1. Settings

We evaluate on METR-LA traffic speeds and AQI36 air quality. All experiments forecast 12 steps from 24 historical steps. For METR-LA, Point and Block-T masks are applied only to the historical input; future targets are evaluated on valid, unmasked raw values. Point masks simulate independent random sensor-value loss, while Block-T masks simulate contiguous temporal outages. AQI36 retains its native validity mask and adds optional Point masks on top. Each condition uses five paired mask seeds, and compared models share the same input masks. The model is trained only with the masked forecasting loss in Eq. (18), without imputation or graph auxiliary losses. HD-TTS-AMP is the main reference. METR-LA also includes BiTGraph connected to the common data, mask, and evaluation interface; because its original training schedule differs from HD-TTS, we interpret it only as a unified-protocol comparison.

Table 1. METR-LA results over five paired mask seeds. Lower is better; the best result within each condition is in bold.

Mask	Rate	Method	MAE	RMSE
Block-T	30%	HD-TTS-AMP	3.2281 ± 0.0108	6.6036 ± 0.0126
		BiTGraph	3.8232 ± 0.0110	7.8839 ± 0.0373
		MR-MSGF	$\mathbf{3.1997 \pm 0.0159}$	$\mathbf{6.5840 \pm 0.0485}$
	50%	HD-TTS-AMP	3.3546 ± 0.0126	6.8780 ± 0.0215
		BiTGraph	3.9729 ± 0.0180	8.1905 ± 0.0458
		MR-MSGF	$\mathbf{3.3079 \pm 0.0195}$	$\mathbf{6.8154 \pm 0.0554}$
	70%	HD-TTS-AMP	3.4650 ± 0.0103	$\mathbf{7.0927 \pm 0.0421}$
		BiTGraph	4.1363 ± 0.0132	8.4812 ± 0.0306
		MR-MSGF	$\mathbf{3.4458 \pm 0.0102}$	7.1088 ± 0.0384
Point	30%	HD-TTS-AMP	3.1048 ± 0.0074	6.3592 ± 0.0154
		BiTGraph	3.6374 ± 0.0124	7.5260 ± 0.0377
		MR-MSGF	$\mathbf{3.0764 \pm 0.0118}$	$\mathbf{6.3091 \pm 0.0447}$
	50%	HD-TTS-AMP	3.1598 ± 0.0112	6.4755 ± 0.0215
		BiTGraph	3.6938 ± 0.0049	7.6577 ± 0.0289
		MR-MSGF	$\mathbf{3.1409 \pm 0.0068}$	$\mathbf{6.4564 \pm 0.0289}$

Table 2. AQI36 cross-domain results over five seeds.

Condition	Method	MAE	RMSE
Native	HD-TTS-AMP	$\mathbf{30.4839 \pm 0.5849}$	$\mathbf{49.3332 \pm 1.4069}$
	MR-MSGF	30.7269 ± 0.2392	49.8879 ± 0.9287
Native+P25	HD-TTS-AMP	31.2034 ± 0.6894	50.3712 ± 1.0346
	MR-MSGF	$\mathbf{31.1169 \pm 0.4161}$	$\mathbf{50.3325 \pm 0.7815}$
Native+P50	HD-TTS-AMP	31.7359 ± 0.4956	$\mathbf{50.4029 \pm 1.2035}$
	MR-MSGF	$\mathbf{31.6451 \pm 0.7093}$	50.5270 ± 1.3859

3.2. Main Results

Table 1 reports METR-LA results. MR-MSGF lowers MAE over HD-TTS-AMP in all tested Block-T and Point settings. It also improves RMSE under Block-T 30%/50% and Point 30%/50%. At Block-T 70%, MR-MSGF still improves MAE but has slightly worse RMSE, indicating that the benefit is not cross-metric consistent under extreme missingness.

3.3. Cross-Domain and Horizon Analysis

AQI36 provides a cross-domain stress test with native missingness. Table 2 shows that MR-MSGF is worse than HD-TTS-AMP under native missingness, improves both metrics with an additional 25% Point mask, and lowers MAE but raises RMSE with an additional 50% Point mask. We therefore treat AQI36 as mixed evidence rather than claiming dataset-independent gains.

We also decompose METR-LA Block-T checkpoints by prediction horizon. At 50% missingness, MR-MSGF lowers both MAE and RMSE at horizons 3, 6, and 12. At 70% missingness, it slightly lowers MAE at all three horizons but raises RMSE at each horizon, matching the aggregate result in Table 1. The extreme-missingness gain is therefore concentrated on average error rather than large-error control.

Table 3. METR-LA Block-T horizon decomposition.

Rate	Horizon	HD-TTS-AMP MAE/RMSE	MR-MSGF MAE/RMSE
50%	3	3.1319/6.2557	**3.0750**/**6.1586**
50%	6	3.3787/6.9274	**3.3333**/**6.8715**
50%	12	3.6532/7.6045	**3.6195**/**7.5840**
70%	3	3.2773/**6.5703**	**3.2520**/6.5811
70%	6	3.4777/**7.1235**	**3.4667**/7.1568
70%	12	3.7302/**7.7343**	**3.7186**/7.7618

3.4. Ablation and Diagnostics

Table 4 isolates the Block-T 50% mechanisms. The full model is best on both metrics. Removing the co-observation score penalty, the reliability gate, or the readout bias all hurts performance, and using only one fine relation source is weaker than the fused graph. This supports the central design choice: co-observation-guided relation fusion and reliability-controlled regional context are complementary.

Table 4. METR-LA Block-T 50% ablation over five paired seeds.

Variant	MAE	RMSE
HD-TTS-AMP	3.3546 ± 0.0126	6.8780 ± 0.0215
Physical only	3.3292 ± 0.0118	6.8605 ± 0.0490
Observation only	3.3348 ± 0.0161	6.8874 ± 0.0587
Fused graph	3.3176 ± 0.0111	6.8288 ± 0.0455
No q_{ij} penalty	3.3149 ± 0.0200	6.8327 ± 0.0547
No rel. gate	3.3136 ± 0.0105	6.8395 ± 0.0420
No readout bias	3.3298 ± 0.0249	6.8512 ± 0.0375
Full MR-MSGF	$\mathbf{3.3079 \pm 0.0195}$	$\mathbf{6.8154 \pm 0.0554}$

As a diagnostic, the Pearson correlation between non-diagonal co-observation rates and physical-edge gates g_{ij} is -0.4051 across 10 full-model checkpoint evaluations. Since larger g_{ij} assigns more weight to the physical graph, this negative association is consistent with greater use of observation-supported relations when co-observation is available, though it is not causal evidence. Table 5 shows that MR-MSGF adds 7.7% parameters while keeping measured inference time at a comparable scale.

Table 5. Post-hoc efficiency statistics.

Method	Params	Test batch time (s)
HD-TTS-AMP	378,351	0.2692 ± 0.0907
MR-MSGF	407,447	0.2672 ± 0.1065

4. CONCLUSION

MR-MSGF forecasts incomplete spatial histories by fusing co-observation-supported and physical relations, then gating regional context by reliability evidence. METR-LA gains validate the design, while 70% Block-T and AQI36 show that it is a useful missingness-aware bias rather than a cross-domain guarantee.

5. REFERENCES

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